

EX. NO: 8

MINI PROJECT I – TOURISM ANALYTICS DASHBOARD USING POWER BI AND TABLEAU

AIM

To design and develop an end-to-end business intelligence solution for a tour operator by retrieving data from a SQL Server data warehouse, preparing and modelling that data, and building an interactive Power BI dashboard and a Tableau dashboard and story that apply the data connection, transformation, modelling, calculation, advanced visualisation and interactivity concepts covered in Unit IV and Unit V.

ALGORITHM

1. Select the Tourism Analytics domain from the five domains provided and define the business problem to be answered.
2. Design a dimensional (star schema) data warehouse consisting of one fact table and six conformed dimension tables.
3. Create the database TourismDW in SQL Server, with a staging schema for the raw extract and a presentation schema for the cleaned model.
4. Load the raw booking extract into the staging table stg.FactTrip_Raw and load all six dimension tables.
5. Profile the raw data in T-SQL to identify duplicate records, invalid values and inconsistent categories.
6. Clean and transform the staged data into dbo.FactTrip by de-duplicating, repairing invalid durations, standardising category values, enforcing referential integrity and deriving additional analytical columns.
7. Apply primary keys, foreign keys and non-clustered indexes so that the star schema relationships are physically enforced in the database.
8. Create analytical views that join the fact table to the dimensions and pre-compute scorecard, trend and customer-value aggregations.
9. Connect Power BI Desktop to SQL Server using a parameterised server name, and import the fact and dimension tables.
10. Apply further cleaning and transformation in Power Query: filter invalid rows, set data types and add custom columns.
11. Build the Power BI data model by defining relationships, hierarchies, calculated columns and DAX measures, including time-intelligence, ranking and what-if measures.
12. Design a six-page Power BI report with KPI cards, advanced visualisations, slicers, drill-down and drill-through, and apply consistent formatting.
13. Export the integrated view from SQL Server as the Tableau extract, and connect Tableau to it.
14. Create calculated fields, table calculations, parameters and filters in Tableau, and build the worksheets that use them.

15. Assemble three interactive Tableau dashboards, add filter and highlight actions, and combine the dashboards into a Tableau story.
16. Interpret the dashboards, record the insights obtained and document the complete solution.

PROGRAM

Step 1: Domain and problem statement

Domain: Tourism Analytics.

IndiaTrails Travel is a tour operator selling domestic and international holiday packages through its own website and app, a call centre, walk-in branches, online travel agencies and partner agents. Bookings, traveller details, destinations, accommodation, transport and pricing are held in an operational system and extracted nightly into a data warehouse. Management can see total sales, but cannot answer the questions that actually drive profit.

The mini project sets out to answer six of them:

- Which destinations generate the most revenue, and how concentrated is that revenue?
- How strongly does demand vary by season, and when should capacity be bought?
- Which booking channels are genuinely profitable once commission is deducted?
- Does the loyalty discount programme pay for itself in extra trips?
- What drives traveller satisfaction, and where is it being eroded?
- How far ahead do travellers book, and does that relate to cancellation?

Step 2: Dataset description

The warehouse follows a star schema: one fact table at the grain of one row per booking, surrounded by six dimension tables.

Table	Type	Rows	Cols	Description
dbo.FactTrip	Fact	12,000	27	One row per booking: costs, revenue, discount, commission, duration, party size, lead time, satisfaction, status
dbo.DimDate	Dimension	2,191	15	Calendar 2022–2027 with year, quarter, month, financial year, season and peak-season flag
dbo.DimDestination	Dimension	32	13	Destination, city, country, region, continent, type, latitude/longitude, cost and popularity index
dbo.DimTraveller	Dimension	2,400	10	Traveller name, gender, age band, nationality, customer segment, loyalty tier, signup date
dbo.DimAccommodation	Dimension	155	7	Property name, type, star rating, price index, average rating
dbo.DimTransport	Dimension	18	6	Transport type, carrier, travel class, cost factor
dbo.DimBookingChannel	Dimension	9	4	Channel name, channel type and commission rate

Table	Type	Rows	Cols	Description
stg.FactTrip_Raw	Staging	12,060	23	The raw extract before cleaning, retained as evidence

Travel dates span 1 January 2023 to 31 December 2026, covering 12,000 clean bookings, 35,635 travellers and ₹296.91 crore of gross revenue.

Step 3: Data source: connecting to SQL Server

The database was created and loaded on a local SQL Server 2022 Express instance. Both Power BI and Tableau connect to the same instance, so the two tools report identical figures.

Property	Value
Server	localhost\SQLEXPRESS
Database	TourismDW
Authentication	Windows Authentication
Power BI connectivity	Import mode, server name held as a Power Query parameter
Tableau connectivity	Live connection using a Custom SQL query on dbo.vw_TripAnalysis

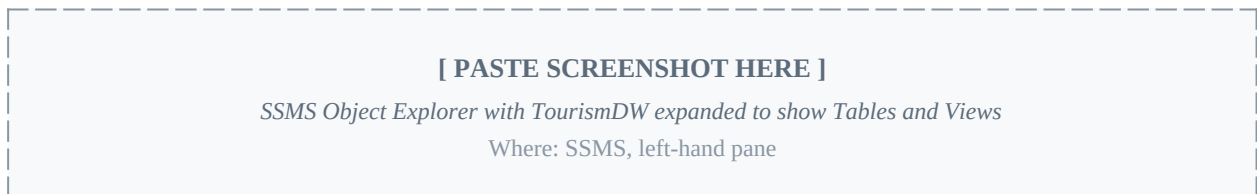


Fig. 8.1 – SQL Server Object Explorer showing the TourismDW database

Step 4: Data preparation: profiling and cleaning

The extract was deliberately treated as untrusted. It was loaded first into stg.FactTrip_Raw and profiled before any use. Profiling found four defects:

Defect found in the raw extract	Count	Cleaning rule applied
Duplicate bookings (same BookingReference)	60	ROW_NUMBER() over BookingReference, keep the lowest TripID
Negative DurationDays (sign-flip entry error)	25	Repair with ABS(), rather than discard the booking
Inconsistent BookingStatus casing — 362 rows stored in upper case	362	UPPER(LEFT(..,1)) + LOWER(SUBSTRING(..,2,50)), then a special case for "No Show". Note: SQL Server's default collation is case-insensitive, so the profile query reports 4 distinct values rather than 8; the defect is still visible in the output, where NO SHOW appears in capitals beside title-case values
Missing SatisfactionScore	2,197	Left as NULL — cancelled and future trips genuinely have no rating, so imputing would distort the average

Rows whose foreign keys did not resolve were excluded by inner-joining every dimension during the load. Four analytical columns were then derived: RevenuePerTraveller, RevenuePerDay, BookingWindow and TripLengthBand.

Stage	Rows
Raw rows in stg.FactTrip_Raw	12,060
Clean rows in dbo.FactTrip	12,000
Rows removed	60

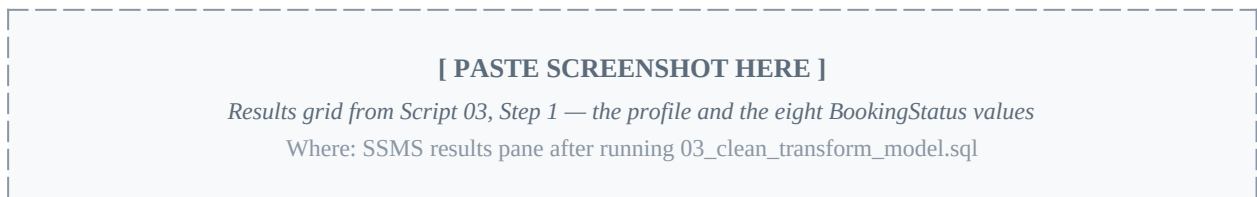


Fig. 8.2 – Raw data quality profile before cleaning

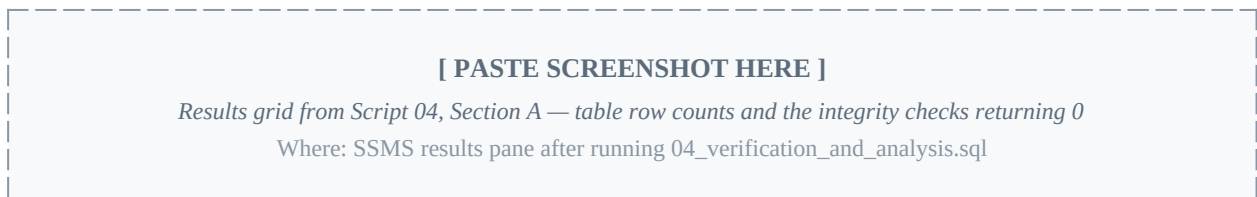


Fig. 8.3 – Row counts and integrity checks after cleaning

Step 5: Data modelling: the star schema

Foreign keys were applied in SQL Server so that the star schema is enforced by the database rather than only assumed by the reporting tools. Non-clustered indexes were added on the four most frequently filtered columns.

Relationship	Cardinality	Notes
FactTrip → DimDate (StartDateKey)	Many-to-one	Active. Drives all time intelligence
FactTrip → DimDate (BookingDateKey)	Many-to-one	Inactive; activated in DAX with USERRELATIONSHIP
FactTrip → DimDate (EndDateKey)	Many-to-one	Inactive; retained for return-date analysis
FactTrip → DimDestination	Many-to-one	Active
FactTrip → DimTraveller	Many-to-one	Active
FactTrip → DimAccommodation	Many-to-one	Active
FactTrip → DimTransport	Many-to-one	Active
FactTrip → DimBookingChannel	Many-to-one	Active
DimAccommodation →	Many-to-one	Inactive. Kept for documentation only: with the

Relationship	Cardinality	Notes
DimDestination		direct FactTrip → DimDestination relationship active, leaving this one active would give Power BI two paths to DimDestination and an ambiguous model

Four analytical views were created on top of the model: vw_TripAnalysis (the fully integrated wide view used by Tableau), vw_DestinationScorecard, vw_MonthlyTrend (with a three-month moving average and a same-month-last-year comparison computed by window functions) and vw_TravellerValue (an NTILE-based customer value segmentation).

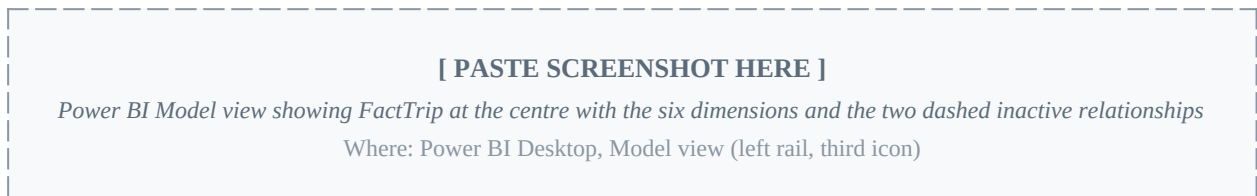


Fig. 8.4 – Star schema in Power BI Model view

Step 6: Power BI: data connection and Power Query transformation

The server name is held as a Power Query parameter so the solution moves between machines without editing any query. Each table is retrieved by navigating the SQL Server catalogue:

```
Source = Sql.Database(ServerName, DatabaseName),  
dbo_FactTrip = Source{[Schema="dbo",Item="FactTrip"]}[Data],
```

A second layer of transformation is applied in Power Query on FactTrip:

- Removed Blank Rows — drops rows with a null TripID or Revenue.
- Filtered Valid Trips — keeps only rows with a positive duration, party size and revenue.
- Changed Type — sets the five monetary and score columns to decimal number.
- Added Satisfaction Band — a conditional column classifying each rating as Promoter, Passive, Detractor or Not Rated.
- Added Discount Pct — DiscountAmount divided by GrossAmount, guarded against division by zero.
- Added Traveller Days — NumTravellers multiplied by DurationDays.

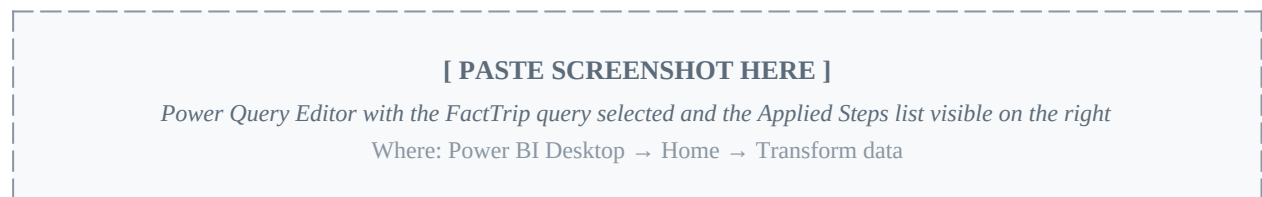


Fig. 8.5 – Power Query Editor showing the applied steps on FactTrip

Step 7: Power BI: calculated columns and DAX measures

The model contains 14 calculated columns and 51 DAX measures organised into nine display folders. The measures below are representative of each category.

Measure	DAX expression	Category
Total Revenue	SUM (FactTrip[Revenue])	Base
Avg Booking Value	DIVIDE ([Total Revenue], [Total Bookings])	Ratio
Cancellation Rate %	DIVIDE ([Cancelled Bookings], [Total Bookings])	Ratio
Net Margin %	DIVIDE ([Net Revenue], [Total Revenue])	Ratio
Revenue YTD	TOTALYTD ([Total Revenue], DimDate[FullDate])	Time intelligence
Revenue PY	CALCULATE ([Total Revenue], SAMEPERIODLASTYEAR (DimDate[FullDate]))	Time intelligence
Revenue YoY %	DIVIDE ([Revenue YoY], [Revenue PY])	Time intelligence
Revenue 3M Moving Avg	AVERAGEX (DATESINPERIOD (DimDate[FullDate], MAX (DimDate[FullDate]), -3, MONTH), CALCULATE ([Total Revenue]))	Time intelligence

Measure	DAX expression	Category
Running Total Revenue	CALCULATE ([Total Revenue], FILTER (ALLSELECTED (DimDate[FullDate]), DimDate[FullDate] <= MAX (DimDate[FullDate])))	Time intelligence
Bookings by Booking Date	CALCULATE ([Total Bookings], USERELATIONSHIP (FactTrip[BookingDateKey], DimDate[DateKey]))	Inactive relationship
Destination Rank	RANKX (ALL (DimDestination[DestinationName]), [Total Revenue], , DESC, DENSE)	Ranking
Pareto Revenue %	Cumulative share using ADDCOLUMNS, FILTER and SUMX over ALLSELECTED destinations	Ranking
Revenue Target	IF (NOT ISBLANK ([Revenue PY]), [Revenue PY] * 1.15)	Target
Target Achievement %	DIVIDE ([Total Revenue], [Revenue Target])	Target
Revenue at Scenario Discount	SUMX (FactTrip, FactTrip[GrossAmount] * (1 - SELECTEDVALUE ('Discount Scenario'[Discount %], 0)))	What-if
Dynamic Title Destination	Concatenates the selected destinations into the visual title	Dynamic title

A what-if parameter table, Discount Scenario, was generated with GENERATESERIES (0, 0.30, 0.01) and bound to a slicer so the whole book can be re-priced at a single discount level.

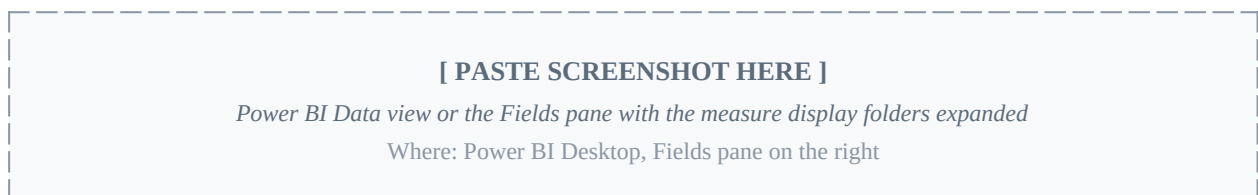


Fig. 8.6 – DAX measures organised in display folders

Step 8: Power BI: dashboard design

The report has six pages and fifty visuals.

Page	Visuals used	Concepts demonstrated
1. Executive Overview	Six KPI cards, line-and-clustered-column combo, donut, bar, treemap, two slicers	KPIs and cards, combo chart, slicers
2. Destination Performance	Map, drill-down column chart on the geography hierarchy, scatter, matrix scorecard	Maps, drill-down, drill-through source, conditional formatting
3. Seasonality & Growth	Month $\times$ year matrix heat map, YoY column chart, ribbon chart, combo chart	Advanced visualisations, time intelligence
4. Customer & Channel	Waterfall, bar, combo, clustered column, table	Waterfall, multi-measure comparison
5. Scenario Planning	What-if slicer, three cards, clustered column, Pareto combo, gauge, funnel, clustered column	What-if parameter, gauge against target, Pareto
6. Destination Detail	Six cards, line chart, bar chart, detail table	Drill-through target page

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The full Executive Overview page with the KPI card strip and all filters cleared

Where: Power BI Desktop, Report view

Fig. 8.7 – Power BI page 1: Executive Overview

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The Destination Performance page, with the drill-down chart expanded to country level

Where: Power BI Desktop, Report view

Fig. 8.8 – Power BI page 2: Destination Performance

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The Seasonality page with the heat map showing background conditional formatting

Where: Power BI Desktop, Report view

Fig. 8.9 – Power BI page 3: Seasonality & Growth

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The Customer & Channel page

Where: Power BI Desktop, Report view

Fig. 8.10 – Power BI page 4: Customer & Channel

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The Scenario Planning page with the discount slicer set to a non-zero value

Where: Power BI Desktop, Report view

Fig. 8.11 – Power BI page 5: Scenario Planning

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The Destination Detail page after right-clicking a destination on page 2 and choosing Drill through

Where: Power BI Desktop, Report view

Fig. 8.12 – Power BI page 6: Destination Detail reached by drill-through

Step 9: Tableau: connection and data integration

The integration itself is performed in SQL Server. The view `dbo.vw_TripAnalysis` joins the fact table to all six dimension tables and exposes a single clean relation of 54 fields, which keeps the join logic in one place and guarantees that Tableau and Power BI report identical figures.

Tableau consumes that view in one of two ways depending on the edition available:

Edition	Connection method	Workbook
Tableau Desktop	Live connection to SQL Server using a Custom SQL query on <code>dbo.vw_TripAnalysis</code>	TourismAnalytics_SQLServer.twb
Tableau Public	The view is exported from SQL Server as a flat extract, which the packaged workbook consumes	TourismAnalytics_Packaged.twb

Tableau Public does not support database connectors — it reads only Excel, text, spatial, statistical and a small number of web sources, with SQL Server, Oracle, MySQL and PostgreSQL restricted to Tableau Desktop. This project was therefore built against the extract route, and script `05_export_extract_for_tableau.sql` is the exact query used to produce that extract from the database:

```
SELECT ... FROM dbo.vw_TripAnalysis ORDER BY TripID;  -- 12,000 rows x 54 columns
```

The extract is therefore a SQL Server output, not an independent data file: all cleaning, conforming and joining happens in the database, and Tableau receives the result. The Tableau Desktop workbook connecting live to the same view is included alongside it for completeness.

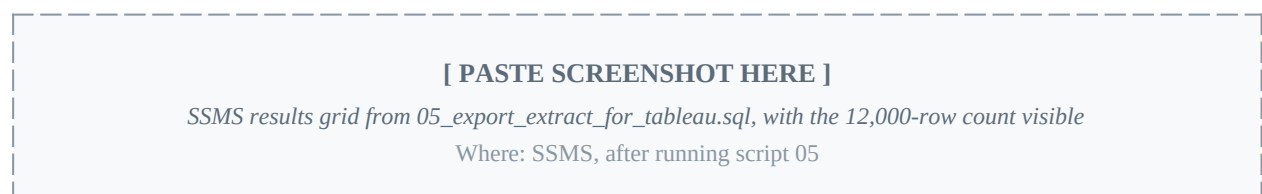


Fig. 8.13 – Exporting the integrated view from SQL Server for Tableau

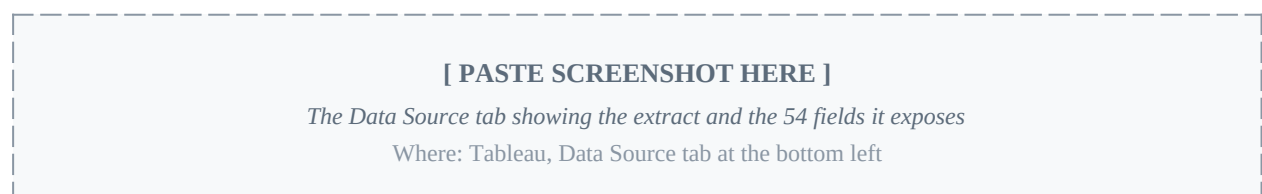


Fig. 8.14 – Tableau Data Source tab

Step 10: Tableau: calculated fields, parameters and filters

Eighteen calculated fields were created, six of which are table calculations, together with four parameters.

Calculated field	Formula	Type
Total Revenue	SUM([Revenue])	Aggregate
Avg Booking Value	SUM([Revenue]) / COUNT([TripID])	Aggregate
Cancellation Rate	SUM(IF [BookingStatus] = "Cancelled" THEN 1 ELSE 0 END) / COUNT([TripID])	Aggregate
Net Margin %	SUM([NetRevenue]) / SUM([Revenue])	Aggregate
Destination Rank	RANK(SUM([Revenue]))	Table calculation
Revenue Moving Avg	WINDOW_AVG(SUM([Revenue]), -2, 0)	Table calculation
Running Revenue %	RUNNING_SUM(SUM([Revenue])) / TOTAL(SUM([Revenue]))	Table calculation
Revenue YoY %	(SUM([Revenue]) - LOOKUP(SUM([Revenue]), -1)) / ABS(LOOKUP(SUM([Revenue]), -1))	Table calculation
Top N Filter	RANK(SUM([Revenue])) <= [Top N Destinations]	Table calc + parameter
Scenario Revenue	SUM([GrossAmount]) * (1 - [Scenario Discount %])	Parameter-driven
Revenue Target	LOOKUP(SUM([Revenue]), -1) * (1 + [Target Growth %])	Parameter-driven
Selected Measure	CASE [Select Measure] WHEN "Revenue" THEN SUM([Revenue]) WHEN "Bookings" THEN COUNT([TripID]) WHEN "Travellers" THEN SUM([NumTravellers]) END	Parameter-driven

Parameter	Data type	Domain	Used by
Top N Destinations	Integer	Range 5 to 25, step 1	Top N Filter, on the Top N Destinations worksheet
Scenario Discount %	Float	Range 0% to 30%, step 1%	Scenario Revenue and Scenario Impact
Target Growth %	Float	Range 0% to 50%, step 5%	Revenue Target
Select Measure	String	Revenue / Bookings / Travellers	Selected Measure, for measure swapping

Four filters — travel year, travel scope, season and channel type — are placed on every worksheet and exposed as interactive controls on the dashboards.

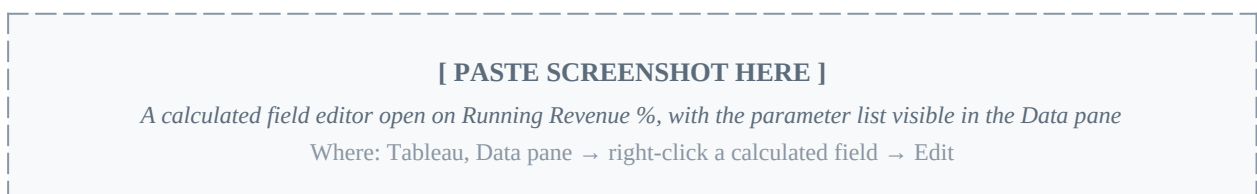


Fig. 8.15 – Calculated field and parameter definitions in Tableau

Step 11: Tableau: worksheets, dashboards and story

Nineteen worksheets were built and assembled into three dashboards and one story.

Dashboard	Worksheets	Interactivity
D1 Executive Overview	Five KPI tiles, revenue trend line, Top N destinations bar, segment treemap, channel profitability bar	Year and travel-scope filter controls, Top N parameter control, highlight action on destination
D2 Destination & Seasonality	Destination map, month $\times$ year heat map, destination Pareto, satisfaction analysis	Filter action from the map to the detail table on D3
D3 Customer & Channel	Traveller demographics, booking window, cost composition, discount scenario, booking detail table	Scenario discount parameter control, channel type and season filter controls

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The complete D1 dashboard with the Top N parameter slider visible
Where: Tableau

Fig. 8.16 – Tableau dashboard D1: Executive Overview

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The complete D2 dashboard with the map and heat map
Where: Tableau

Fig. 8.17 – Tableau dashboard D2: Destination & Seasonality

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The complete D3 dashboard with the discount parameter set to a non-zero value
Where: Tableau

Fig. 8.18 – Tableau dashboard D3: Customer & Channel

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The story with all four story points visible along the navigator strip
Where: Tableau

Fig. 8.19 – Tableau story

Step 12: Insights obtained

Insight 1 — Revenue is growing, but the growth is almost entirely international.

Travel scope	Bookings	Revenue (₹ Cr)	Avg booking value	Share of revenue
International	8,731	270.39	₹3,09,693	91.1%
Domestic	3,269	26.51	₹81,101	8.9%

International trips are 72.8% of bookings but 91.1% of revenue, because the average international booking is worth 3.8 times the average domestic one. International revenue grew every year (+18.1%, +13.1%, +7.0%); domestic revenue fell 4.1% in 2026 after a strong 2025. Total revenue rose 43% between 2023 and 2026.

Insight 2 — Demand swings by a factor of 2.2 across the year.

Month	Revenue (₹ Cr)	Seasonality index
December	37.07	150
October	34.13	138
May	31.62	128
April	31.16	126
November	29.91	121
June (lowest)	16.83	68

December earns 2.2 times what June earns. The peak months — April, May, October, November and December — carry a further 18% price premium, which means capacity bought in advance for those months is where margin is won or lost.

Insight 3 — Online travel agencies buy volume at a real cost to margin.

Channel	Type	Bookings	Revenue (₹ Cr)	Commission (₹ L)	Net margin
MakeMyTrip	OTA	1,956	49.92	698.94	86.0%
Booking.com	OTA	1,785	42.92	643.87	85.0%
Expedia	OTA	1,481	37.34	597.49	84.0%
Company Website	Direct	1,582	39.01	78.02	98.0%
Mobile App	Direct	1,431	35.35	70.70	98.0%
Walk-in Branch	Offline	621	14.86	0.00	100.0%

The three OTAs together bring 43% of bookings but surrender ₹19.4 crore in commission. MakeMyTrip is the largest channel by gross revenue, yet the company website delivers more net revenue than Booking.com on fewer bookings. Shifting even a tenth of OTA volume to the direct channels would be worth roughly ₹1.9 crore a year at current rates.

Insight 4 — The loyalty discount is not buying additional trips.

Loyalty tier	Travellers	Trips	Trips per traveller	Avg discount	Avg trip value
Blue	425	2,141	5.04	3.27%	₹2,85,138
Silver	605	3,050	5.04	6.16%	₹2,35,807
Gold	546	2,766	5.07	9.40%	₹2,40,538
Platinum	809	4,043	5.00	13.43%	₹2,40,918

Trips per traveller is essentially flat at about 5.0 across all four tiers, while the average discount rises from 3.3% to 13.4%. The programme is therefore transferring margin to the most loyal travellers without generating extra bookings from them. This is the clearest commercial finding in the dataset and would justify a controlled test of a lower Platinum discount.

Insight 5 — Peak season costs half a satisfaction point at every quality level.

Star rating	Peak season score	Off-peak score	Difference
5 star	6.80	7.32	-0.52
4 star	6.35	6.90	-0.55
3 star	5.91	6.49	-0.58
2 star	5.50	6.03	-0.53

The penalty is almost identical at every star rating, which points to crowding and service strain during peak months rather than to any particular grade of property. Upgrading accommodation would not fix it; managing peak-season capacity would.

Insight 6 — Booking earlier does not reduce cancellation.

Booking window	Bookings	Avg booking value	Cancellation rate
Last minute (0–7 days)	347	₹2,63,886	6.05%
Short (8–30 days)	3,752	₹2,46,831	6.50%
Medium (31–90 days)	6,609	₹2,46,280	7.13%
Early bird (90+ days)	1,292	₹2,50,548	6.81%

Cancellation sits in a narrow 6.1% to 7.1% band regardless of how far ahead the booking was made, so lead time is not a useful predictor of cancellation risk here. Last-minute bookings are also the most valuable on average, which argues against discounting late inventory too aggressively.

Insight 7 — Revenue is broadly spread, not concentrated.

The top ten destinations account for 48.4% of revenue across 32 destinations, and no single destination exceeds 5.8%. London Explorer leads at ₹17.08 crore (5.75%), followed by Bangkok & Phuket at ₹14.70 crore and Maldives Atolls at ₹14.60 crore. This is a healthier distribution than the classic 80/20 pattern and means no single destination failing would threaten the business — but it also means there is no obvious place to concentrate marketing spend.

Transport is the largest single cost component at 46.8% of gross booking value, ahead of accommodation at 28.6% and activities at 24.6%, which reflects how international the mix is.

RESULT

Thus a complete Tourism Analytics business intelligence solution was designed and developed. Data was retrieved from a SQL Server data warehouse, profiled, cleaned and modelled into a star schema; an interactive six-page Power BI dashboard with 51 DAX measures, 14 calculated columns, nine relationships and a what-if parameter was built; a Tableau workbook with 19 worksheets, 18 calculated fields, 4 parameters, 3 dashboards and a story was built on an extract of the same integrated view; and the resulting dashboards were interpreted to produce seven business insights. The outputs were verified successfully.